

# Seminar 1. Bayesian Reasoning

## Bayesian Methods

1. A medical test finds a serious disease in a patient. The test is 99% accurate: if the patient has the disease, the test is positive with probability 99%; if the patient does not have the disease, the test is negative with probability 99%. The disease is rare: only one person in 10 000 has it. Find the probability that the patient really has the disease.
2. Consider the following probabilistic model. At a seminar, a student answers badly ( $A = 1$ ) or well ( $A = 0$ ). This depends on whether the student has depression ( $D = 1$  or  $D = 0$ ) and whether the student went to a party the night before ( $P = 1$  or  $P = 0$ ). The party can also give the student a headache ( $H = 1$ ). A bad answer makes the teacher unhappy ( $T = 1$ ). Figure 1 shows the causal relations in the model. The probabilities are:

| $p(A = 1 \mid P, D) \mid P \quad D$ | $p(H = 1 \mid P) \mid P$ | $p(T = 1 \mid A) \mid A$ |
|-------------------------------------|--------------------------|--------------------------|
| 0.999                               | 0.9                      | 0.95                     |
| 0.9                                 | 0.2                      | 0.5                      |
| 0.9                                 |                          |                          |
| 0.01                                |                          |                          |

$$p(P = 1) = 0.2, \quad p(D = 1) = 0.4.$$

Find  $p(P = 1 \mid H = 1)$ ,  $p(P = 1 \mid T = 1)$ , and  $p(P = 1 \mid T = 1, H = 1)$ .

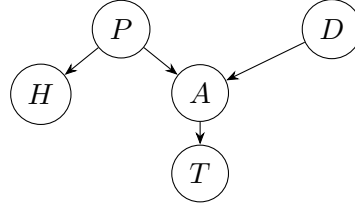


Figure 1: Causal relations for Problem 2

3. Let  $x_1 \sim \text{Poiss}(\lambda_1)$  and  $x_2 \sim \text{Poiss}(\lambda_2)$  be two independent Poisson random variables, that is,  $p(x_i = k) = \exp(-\lambda_i) \frac{\lambda_i^k}{k!}$ ,  $k = 0, 1, 2, \dots$ . Prove that  $x_1 + x_2 \sim \text{Poiss}(\lambda_1 + \lambda_2)$ .
4. Let  $x_1, x_2, \dots, x_N$  be an independent sample from  $\text{Poiss}(\lambda)$ . Find the maximum likelihood estimate of  $\lambda$ .
5. Let  $x_1, x_2, \dots, x_N$  be an independent sample from  $\text{Poiss}(\lambda)$ . The prior distribution is the gamma distribution

$$p(\lambda) = \mathcal{G}(\lambda \mid a, b) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} \exp(-b\lambda), \quad a, b > 0.$$

Find the Bayesian estimate of  $\lambda$  as the mean of the posterior distribution  $p(\lambda \mid x_1, \dots, x_N)$ .

### Notes on comparing the estimates from Problems 4 and 5.

1. For more about non-informative priors, see Section 5.4 “Priors” in Kevin P. Murphy, *Machine Learning: A Probabilistic Perspective*.
2. The prior  $p(\lambda) \propto \frac{1}{\lambda}$  is a uniform prior on  $\log \lambda$ . To see this, use the change-of-variables rule for the density of a function of a random variable. See, for example, [this article](#).